

Tic-Tac-Total

AQA A-Level Computer Science NEA

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## Contents

## Section A: Analysis

### Context Of Problem

Mathematics is a compulsory subject within all educational institutions in the United Kingdom up to the age of 16, where students are required to complete General Certificate of Secondary Education (GCSE) examinations. As a core component of STEM (Science, Technology, Engineering, and Mathematics), it plays a critical role in both academic progression and everyday life. Mathematical skills are essential for practical tasks such as budgeting finances, interpreting data, measuring quantities, and solving real-life problems. Mathematics is assessed through a range of examination boards, each with differing structures and assessment styles.

Nationally, approximately 32% of students fail mathematics examinations each year across GCSE and A-Level qualifications, highlighting a significant and persistent issue within mathematics education. A key contributing factor is the difficulty many students experience in keeping pace with lessons, which can lead to disengagement and reduced confidence over time. At A-Level, students commonly struggle with complex topics such as integration and differentiation, which require both accuracy and strong procedural knowledge. At GCSE level, students frequently encounter difficulties in algebraic manipulation, problem-solving, and interpreting worded questions.

This issue can be further understood through specific institutional data. Harington School, a relatively small sixth form centre with approximately 300 students, offers both Mathematics and Further Mathematics at A-Level under the Pearson Edexcel specification. In 2025, the average mathematics grade achieved was a B, and 82.2% of all grades across the school were at grade C or above. While this indicates generally strong academic performance, it also suggests that a proportion of students are not achieving the highest grades, particularly in more demanding areas of the subject.



In contrast, Stamford Welland Academy, with approximately 600 students aged 11–16, demonstrates more significant challenges at GCSE level. In 2025, only 18.1% of students achieved a Grade 5 or above in both Mathematics and English, which is substantially below the local authority average of 41.9%. This indicates a clear attainment gap and suggests that many students struggle to meet expected standards. The school has implemented interventions such as “Session 6,” an after-school support programme designed to improve attainment; however, the available data suggests that its impact has been limited.



Overall, both institutions demonstrate that, despite differing levels of attainment, mathematics remains a challenging subject for many students. The combination of national statistics and local performance data highlights the need for more effective and engaging methods to support students in developing their mathematical skills.

## Analysis of Problem

Mathematics, as a core subject, is expected to be understood at a fundamental level by all students; however, a range of systemic, psychological, and educational challenges prevent this from being achieved. One significant structural issue is the requirement for students who fail mathematics to resit examinations. This can create a cycle of repeated failure, often referred to as a “resit crisis,” where students are required to retake the same assessments without sufficient variation in approach or support, leading to further disengagement.

Psychological factors also play a major role in hindering mathematical performance. Many students experience high levels of anxiety when approaching mathematics, often due to pressure to produce correct answers within limited time constraints. This can negatively impact performance, particularly in exam situations where time efficiency is essential. Additionally, repeated difficulty can lead to a loss of confidence, causing students to disengage and avoid practising mathematics altogether.

Educational approaches further contribute to the problem. Traditional teaching methods often rely heavily on repetitive practice and standardised question formats, which may not sustain student engagement over time. In addition, modern classroom practices can sometimes involve students passively copying notes from a screen or presentation, which limits active participation and reduces opportunities for deeper engagement with the content. As a result, students may struggle to remain motivated, particularly when working independently outside of lessons. Furthermore, many students find it difficult to effectively revise mathematics in their own time, as existing methods may lack interactivity or immediate feedback.

A common challenge is the difficulty students face with worded problems, which require them to interpret information and apply appropriate methods under time pressure. This highlights the importance of not only practising mathematical techniques but also developing the ability to apply them efficiently in exam-style conditions.

Overall, these issues demonstrate a need for a more engaging and accessible approach to mathematics practice—one that supports independent learning, reinforces skills through active participation, and helps students improve both confidence and performance under timed conditions.

## Interview

To better understand the needs of the intended users and ensure the proposed solution is appropriate, interviews were conducted with two students: one studying Mathematics at A-Level and one studying Mathematics at GCSE. These interviews aimed to identify common difficulties, attitudes towards mathematics, and preferences for revision methods. The findings will be used to inform the key features and design of the solution.

### Interview 1: A-Level Mathematics Student

The first interview was conducted with an A-Level Mathematics student currently studying under the Pearson Edexcel specification. This student has prior experience of GCSE mathematics and is familiar with the increased difficulty and expectations at A-Level.

#### **Q1: How do you find A-Level Mathematics compared to GCSE?**

**A:** It is quite a big step up. At GCSE, you can often follow methods quite easily, but at A-Level you actually need to understand why things work rather than just memorising them. If you don't properly understand a topic, it becomes very difficult to answer unfamiliar questions.

#### **Q2: Which areas of mathematics do you find most challenging?**

**A:** Integration and differentiation, mainly because there are lots of different techniques. It's easy to forget small steps or apply the wrong method, especially if you haven't practised regularly.

#### **Q3: How do you currently revise mathematics?**

**A:** I mostly use past papers and textbook questions. Sometimes I go over my notes, but I don't always find that helpful unless I already understand the topic.

#### **Q4: Do you find this method effective?**

**A:** It works to an extent, but it gets very repetitive. After a while, it feels like I'm just doing the same type of questions over and over again without really improving much. It can also be quite boring, so I don't always stay focused for long periods of time.

#### **Q5: How confident do you feel when answering questions under timed conditions?**

**A:** Not very confident. In class I can usually complete questions, but in exams I feel rushed and more likely to make mistakes, especially on longer questions.

#### **Q6: What do you think causes this lack of confidence?**

**A:** Probably a combination of not practising enough under time pressure and not always being fully secure on the methods. Most of the time we practise questions straight after learning the topic, rather than coming back to them later in a timed fashion.

#### **Q7: Do you enjoy learning mathematics? Why or why not?**

**A:** Sometimes, especially when I understand the topic and can answer questions correctly. But when I struggle, it becomes frustrating quite quickly. Lessons can also feel repetitive at times, which makes it harder to stay motivated.

**Q8: Would a more interactive or game-based approach to revision help you?**

**A:** Yes, definitely. I think it would make revision more engaging and less of a chore. If it felt more like a game, I would probably be more likely to use it regularly.

**Q9: What features would you like in a revision tool?**

**A:** Quick-fire questions, something that tracks progress over time, and possibly a competitive element, like playing against someone else or trying to beat a score. That would make it more interesting.

**Q10: Would timed challenges or competition improve your exam performance?**

**A:** Yes, because it would help me get used to working under pressure. Competing against someone else or trying to beat a score or timer would probably push me to think faster, and over time I would become both quicker and more accurate when answering questions.

**Interview 2: GCSE Mathematics Student**

The second interview was conducted with a GCSE Mathematics student studying under the AQA specification. This student represents a younger user group who may still be developing both confidence and core mathematical skills.

**Q1: How do you find mathematics as a subject?**

**A:** It's quite difficult. Sometimes I understand it in class, but when I try questions on my own, I get stuck quite quickly and don't always know what to do next.

**Q2: What topics do you struggle with most?**

**A:** Algebra, especially because it comes up a lot in higher-level questions. I also struggle with worded questions because I don't always understand what the question is asking me to do.

**Q3: What usually happens when you don't understand a topic in class?**

**A:** I tend to fall behind, and then it becomes harder to keep up with the next lessons because they build on what we've already learned. After that, I just lose confidence.

**Q4: How do you revise mathematics outside of lessons?**

**A:** I don't really revise it much unless I have homework or a test coming up. When I do revise, it's usually just answering questions from a worksheet or textbook.

**Q5: Why do you not revise it regularly?**

**A:** Mainly because it feels boring, and I'm not always sure what I should be revising or where to start. I also find it harder to stay focused when I'm working on my own, and sometimes it just feels pointless, like you either understand maths or you don't.

**Q6: Do you ever revise with other people?**

**A:** Not usually, but I think it would help. If I was revising with someone else or competing against them, I would probably be more motivated to actually do it.

**Q7: Do you find lessons engaging?**

**A:** Sometimes, but quite often it's just copying notes from the board or doing similar questions, which can get repetitive after a while and doesn't really help me stay interested.

**Q8: Would you be more likely to revise if it was more interactive or game-based?**

**A:** Yes, definitely. If it was more like a game or something competitive, I would be much more willing to revise. It would make it feel less like work and more enjoyable.

**Q9: Would instant feedback on answers help you?**

**A:** Yes, because then I would know straight away if I got something wrong and could try again. At the moment, I sometimes don't find out until later, which isn't as helpful.

From these interviews, several key themes have emerged. Both students identified a lack of engagement as a major issue when revising mathematics, with traditional methods such as completing past papers or copying notes often described as repetitive and unmotivating. This indicates that any effective solution should incorporate interactive elements that encourage active participation, rather than passive learning, in order to maintain user interest over time.

In addition, both students highlighted a lack of confidence, particularly when working under timed conditions or when revisiting more complex topics. This suggests that a suitable solution should provide opportunities for practising mathematics in a way that reflects exam-style pressure, allowing users to gradually build both speed and accuracy in a low-risk environment.

Another important finding is that students struggle to maintain motivation when revising independently, particularly outside of structured lesson time. However, both students indicated that incorporating competition or working alongside others would make revision more engaging. This demonstrates the need for a solution that can be accessed easily outside of school, college, or sixth form, while also encouraging competitive interaction to increase motivation and sustain engagement.

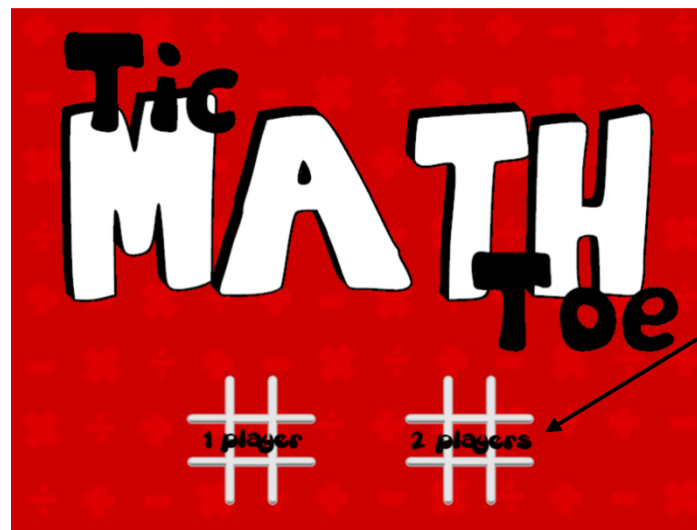
Furthermore, both students expressed a clear preference for a more engaging and interactive approach, with specific interest in game-based and competitive elements. This suggests that incorporating features such as progression tracking and competitive gameplay could encourage more consistent use and increase motivation to practise regularly.

Finally, the importance of immediate feedback was highlighted, particularly by the GCSE student. This indicates that users benefit from being able to quickly identify and learn from mistakes, reinforcing understanding more effectively than delayed feedback.

It can therefore be concluded that any solution to this problem would need to be interactive, engaging, and easily accessible beyond the classroom environment. It should support consistent practice through competitive gameplay, provide timed questions to reflect exam conditions, and include immediate feedback to reinforce learning and improve overall performance.

## Existing Systems

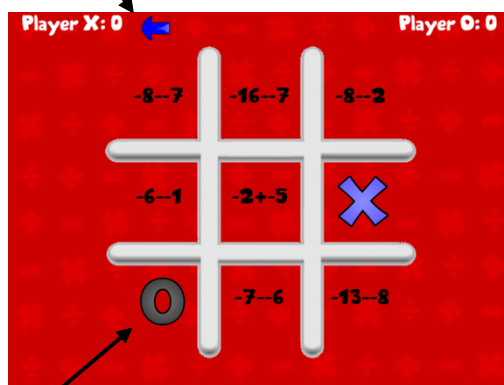
Tic Tac Toe Math – RoomRecess.com



Clear indication of whose turn it is

Allows for single player, but more importantly two player use.

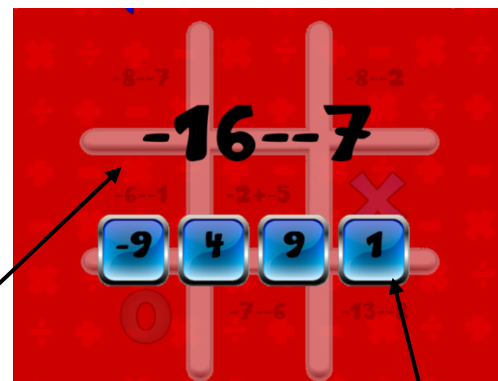
- Competitive nature
- Having a study buddy.



Symbols placed in grid once a question is answered correctly.

No symbol placed if incorrect answer given, and then player turn switched.

A question is enlarged, so it is illustrated clearly to the user that it is for them to answer, once a tile space is clicked.



Questions are all multiple choice, assisting.

This feature can be removed in order to add the extra bit of challenge, which in turn will help users learn and retain learning better and longer.

Keeps track of basic statistics like how many times a player has won in the session

**Player O: 1**

## MathsBuster

**Algebra**

Go to: [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#) [10](#) [11](#)

**Question 1** (2887) [Show a worked example](#) [Show revision topic](#)

Make the given letter the subject of each formula below.  
Give any fractions in your answer in their simplest form.

a. Make  $x$  the subject of  $y = 9x + 2$

$x = \frac{y-2}{9}$  ✓

b. Make  $z$  the subject of  $w = 2 - 9z$

$z = \frac{w+2}{9}$  ✗  $\frac{2-w}{9}$

☹ Not quite. Find out where you went wrong and try again  
(Click a cross to see the correct answer.)

Clearly states the topic of each of the question, so the user can identify subject area that they are good at, so won't need much more practice, and ones that need work on.

Instant feedback which clearly displays if you get the right or wrong answer

Provides the correct answer for the user to learn from, if their answer was incorrect

**⌚ Timed Tests**

These tests are really important for **MathsBuster** to work for you!

- Try Timed Test 1 (all three parts) first. Each part takes about 30 minutes.
- After each part MathsBuster works out which topics you're weaker at and suggests what you should revise. When you've finished a whole Test, you'll get an approximate **working grade**.
- If you've done well in a Test you might have **unlocked some more levels** in Levelled Practice – [take a look](#).
- When you've improved on all your weak topics from the first Test, take Timed Test 2, 3 or 4. They test you on different topics from Test 1 so they'll uncover more weak topics.

This system also has a section dedicated to timed questions, for users to be able to practice in an exam style environment.

Again, quick feedback on the level that a user is working at which ...

## Identification of Users

The intended users of this system are GCSE and A-Level Mathematics students, typically aged between 14 and 18, with a range of abilities. The system is targeted towards, but not limited to, students who lack confidence, struggle with engagement, or find traditional revision methods repetitive and ineffective, as well as those aiming to improve their overall performance in mathematics. The system also targets the large proportion of candidates who find timed exam conditions challenging. As a result, it must incorporate timed questions to improve both speed and accuracy in a controlled and low-pressure environment, as well as interactive elements specifically designed to increase engagement and encourage consistent use, without overwhelming the user.

In terms of technical ability, users of this age group are generally familiar with digital devices and common interface elements such as menus and buttons. However, the system must remain intuitive and easy to navigate so that users can focus on answering questions rather than learning how to use the program. This means the interface should be clear, responsive, and require minimal instruction.

The system will be used on personal devices such as laptops or desktop computers, both in school and at home. As it is not intended for use on large display screens such as teacher interactive whiteboards, interface elements such as text, buttons, and interactive components (e.g. sliders and dialogue pop-ups) should be appropriately sized for individual screens, ensuring clarity and ease of interaction without requiring excessive scaling.

Additionally, the system is intended for two-player use, meaning that interaction between users is a key component. Therefore, the design must clearly display game states, player turns, and outcomes in an easily understandable way to avoid confusion during gameplay. Visual clarity is particularly important to ensure that both players can quickly interpret the current state of the game and respond accordingly.

Finally, given that users may have differing levels of mathematical confidence, the system must also accommodate a range of abilities. This can be achieved through clear question presentation, appropriate difficulty selection (GCSE or A-Level), and immediate feedback to support learning. The inclusion of timed elements must also be carefully balanced to provide challenge without overwhelming the user, ensuring that it supports improvement rather than discouraging participation.

## REQUIREMENTS

From the interview it is clear that the product needs ... :

- Give instant feedback (correct or incorrect)
- Include maths questions from a range of different topics, including the ones students typically find challenging (integration, differentiation, algebra)
- Should include questions under timed conditions in some way, in order to embody exam conditions
- Include an aspect of randomness to keep the game engaging, unlike lessons which has been described to be boring and repetitive
- To also keep things interesting and engaging introduce a form of competition, most likely pvp (2 players) to also include revising with a buddy, which is encouraging.
- Be able for users to access and use the program wherever, and not just in school or 6<sup>th</sup> form
- The program should be interactive and game based

From the identification of Users ... :

From the exploration of current systems that currently exist for students to revise maths ...

Acceptable Limitations

Success Criteria

## Section B: Documented Design

To address the identified problems, I intend to develop an interactive software application that reinforces mathematical skills through an engaging and gamified approach. The program will be based on an enhanced version of the widely recognised game Tic-Tac-Toe, adapted to incorporate educational elements. The core gameplay will remain familiar: players aim to place their symbols in a consecutive line across a grid of variable size in order to win.

However, progression within the game will be dependent on mathematical ability. Before placing a symbol, a player must correctly answer a mathematics question, both worded and non-worded. If the answer is correct, the symbol is placed as intended; if incorrect, the move is forfeited and control passes to the opposing player. This mechanic ensures that success in the game is directly linked to mathematical understanding, encouraging active learning and revision.

To further enhance engagement, additional features will be implemented. Special “bomb tiles” will be randomly distributed across the grid and hidden from players. When triggered, a bomb tile will remove any symbols within a surrounding 3×3 area, resetting those tiles and introducing an element of unpredictability. Players will have the opportunity to defuse a bomb by correctly answering a more difficult, timed question. This feature is designed to simulate exam conditions, encouraging quick thinking and reinforcing knowledge under pressure. If successfully defused, the player’s symbol will be placed without triggering the bomb’s effects.

The program will also include a reset function, allowing the game to return to its initial state, including clearing the grid, redistributing bomb tiles, randomly selecting the starting player, and resetting win counters. A stat-tracking system will record the number of games won by each player, the highest win streak of each player in the current session and number of questions answered correctly and incorrectly, promoting a sense of competition and sustained engagement. Overall, this solution aims to combine educational content with interactive gameplay to create a more motivating and effective learning experience for students.

## Section C: Technical Solution

### Final Product

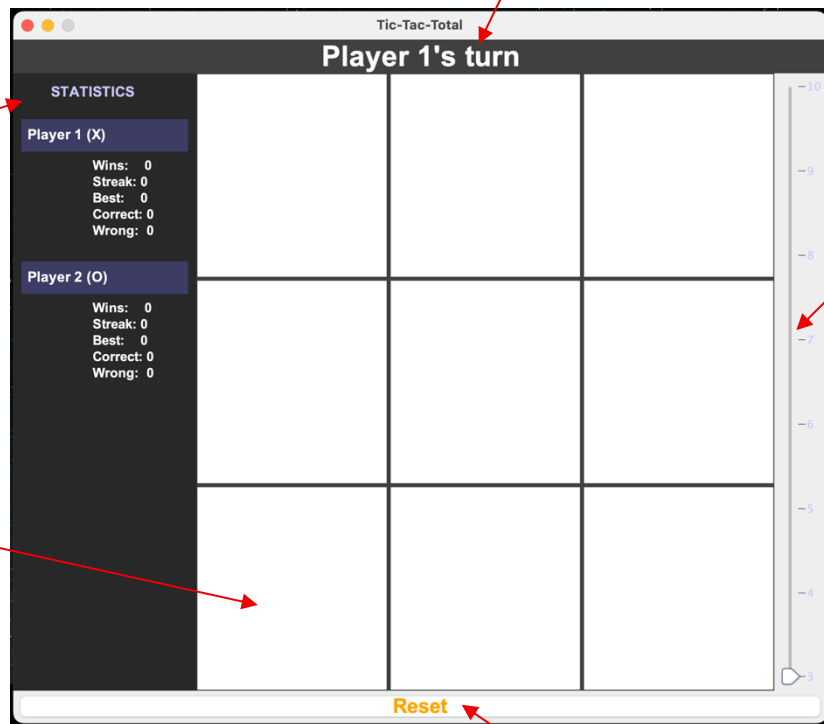
Throughout the duration of the game, from the start, this label displays which players turn it is currently.

Throughout the duration of the session, this panel accurately displays each players statistics for this session. Including:

- Total number of wins
- Current win streak
- Highest win streak
- Total number of correctly answered questions
- Total number of incorrectly answered questions

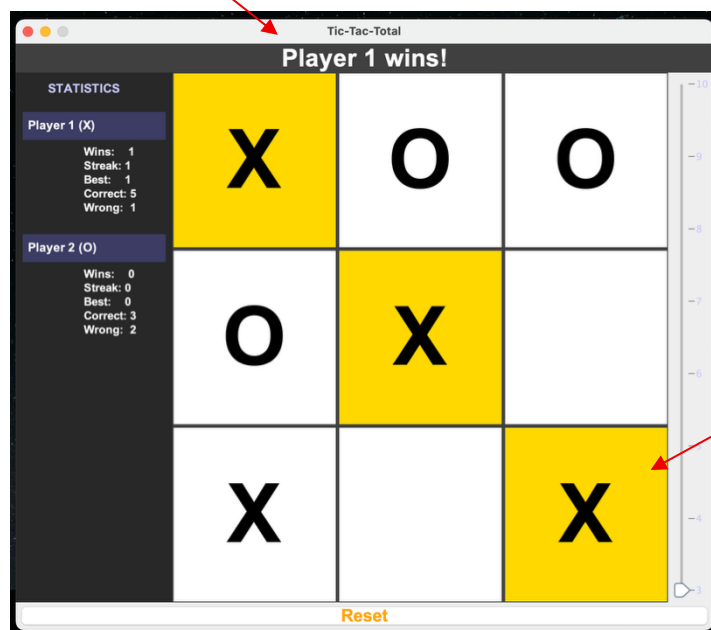
Each tile is represented as button a button. Appropriately displayed in it corresponding number to the Jslider (e.g., 3x3).

When clicked this button places the current player's symbol in the tile and then disabled the opportunity for the button to be clicked again and subsequently unable to change the symbol.



When moved up and down the slider, the following occurs:

- The newly appropriate grid size is displayed
- The game is reset (all symbols removed) and new game is started (which included its own calls such as selecting a new random starting player)



When clicked this button resets the game back to its original starting state. It moves all symbols from the tiles and selects a new random starting player. It can be clicked at any time.

